

SOP-WTR-3600: Boiler Feedwater Chemistry and Treatment

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Section 1. Purpose and Scope

Governs feedwater and boiler water chemistry limits and treatment for steam generation equipment.

Section 2. Monitoring Frequency

Clause 2.1 — Feedwater and boiler water samples (pH, dissolved oxygen, conductivity, phosphate residual as applicable) are tested at least once per shift for high-pressure boilers, daily for low-pressure boilers.

Section 3. Chemistry Limits

Clause 3.1 — Boiler water chemistry shall be maintained within the limits specified on the boiler manufacturer's chemistry guideline for its operating pressure; operating outside limits for more than 2 consecutive sample periods requires notification to the Utilities Department shift supervisor.

Clause 3.2 — Dissolved oxygen in feedwater exceeding the guideline limit requires immediate deaerator/chemical oxygen scavenger system check — sustained excursion risks pitting corrosion damage that is not readily reversible.

Section 4. Blowdown

Clause 4.1 — Continuous and bottom blowdown rates are adjusted to maintain chemistry within limits; bottom blowdown is performed at minimum once per shift regardless of chemistry results, to control sludge accumulation.

Section 5. Records

Clause 5.1 — Water chemistry logs are filed to the Utilities File and retained for 3 years.